



Cloud Watch Alarms Assignment

MSc(computer Science) (Savitribai Phule Pune University)



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CloudWatch Alarms Assignment

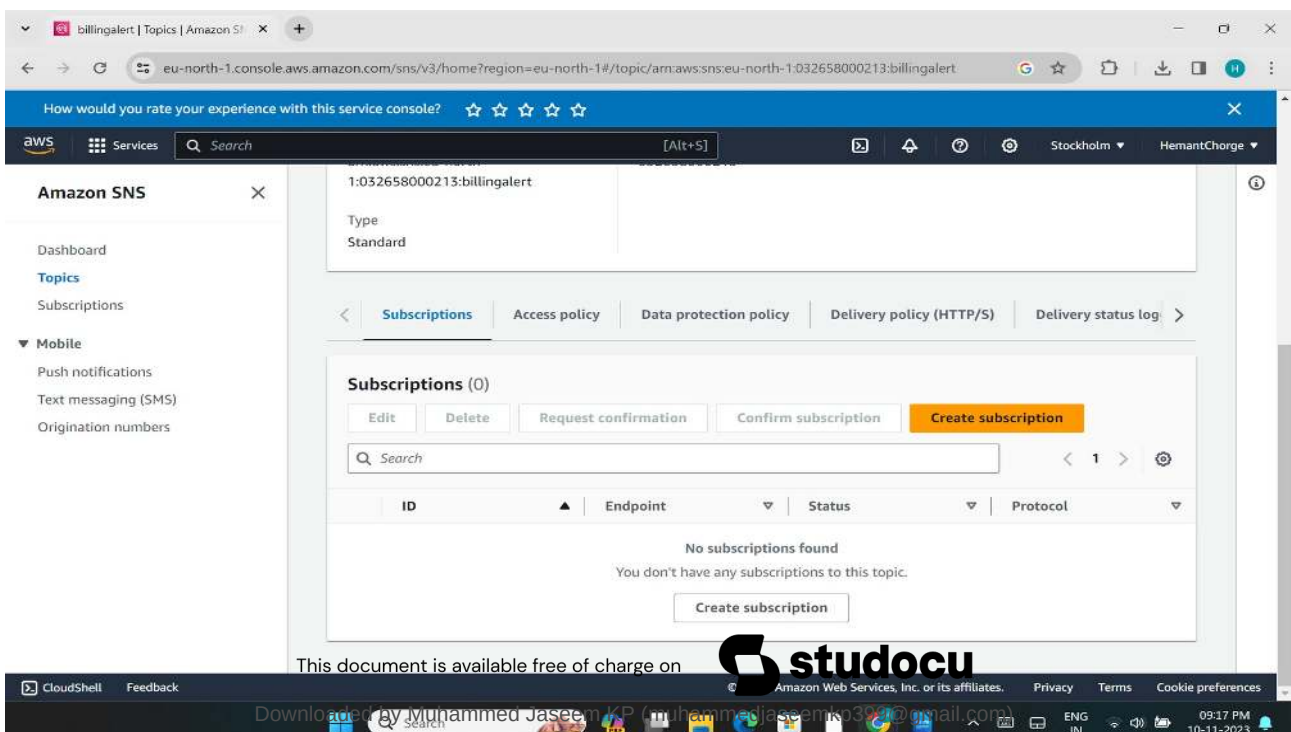
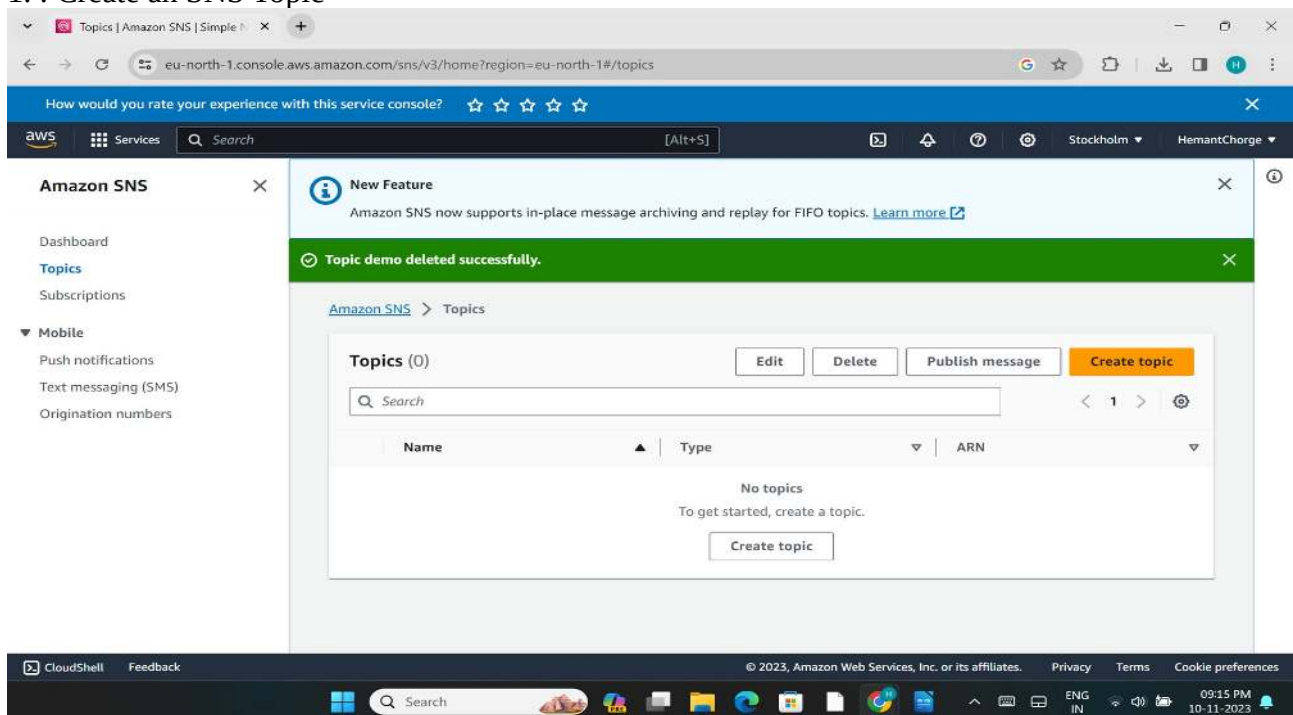
Problem Statement: You work for XYZ Corporation. To maintain the security of the AWS account and the resources you have been asked to implement a solution that can help easily recognize and monitor the different users. Also, you will be monitoring the machines created by these users for any errors or misconfigurations.

Tasks To Be Performed: 1. Create a CloudWatch billing alarm which goes off when the estimated charges go above \$500.

2. Create a CloudWatch alarm which goes off to an Alarm state when the CPU utilization of an EC2 instance goes above 65%. Also add an SNS topic so that it notifies the person when the threshold is crossed.

Solution:

1. : Create an SNS Topic



Create subscription | Subscripti

eu-north-1.console.aws.amazon.com/sns/v3/home?region=eu-north-1#/create-subscription

How would you rate your experience with this service console? ☆☆☆☆

aws Services Search [Alt+S]

Protocol
The type of endpoint to subscribe:
Email

Endpoint
An email address that can receive notifications from Amazon SNS.
hemantchorge45@outlook.com

After your subscription is created, you must confirm it. Info

Subscription filter policy - optional Info
This policy filters the messages that a subscriber receives.

Redrive policy (dead-letter queue) - optional Info
Send undeliverable messages to a dead-letter queue.

Cancel Create subscription

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Search

ENG IN 09:18 PM 10-11-2023

Subscription: 1ba5b692-fbec-4

eu-north-1.console.aws.amazon.com/sns/v3/home?region=eu-north-1#/subscription/arn:aws:sns:eu-north-1:032658000213:billingalert...

How would you rate your experience with this service console? ☆☆☆☆

aws Services Search [Alt+S]

Amazon SNS

Dashboard
Topics
Subscriptions

Mobile
Push notifications
Text messaging (SMS)
Origination numbers

Amazon SNS > Topics > billingalert > Subscription: 1ba5b692-fbec-4509-b1b5-070bdc95d3af

Subscription: 1ba5b692-fbec-4509-b1b5-070bdc95d3af

Edit Delete

Details

ARN
arn:aws:sns:eu-north-1:032658000213:billingalert:1ba5b692-fbec-4509-b1b5-070bdc95d3af

Status
Pending confirmation

Endpoint
hemantchorge45@outlook.com

Protocol
EMAIL

Topic
billingalert

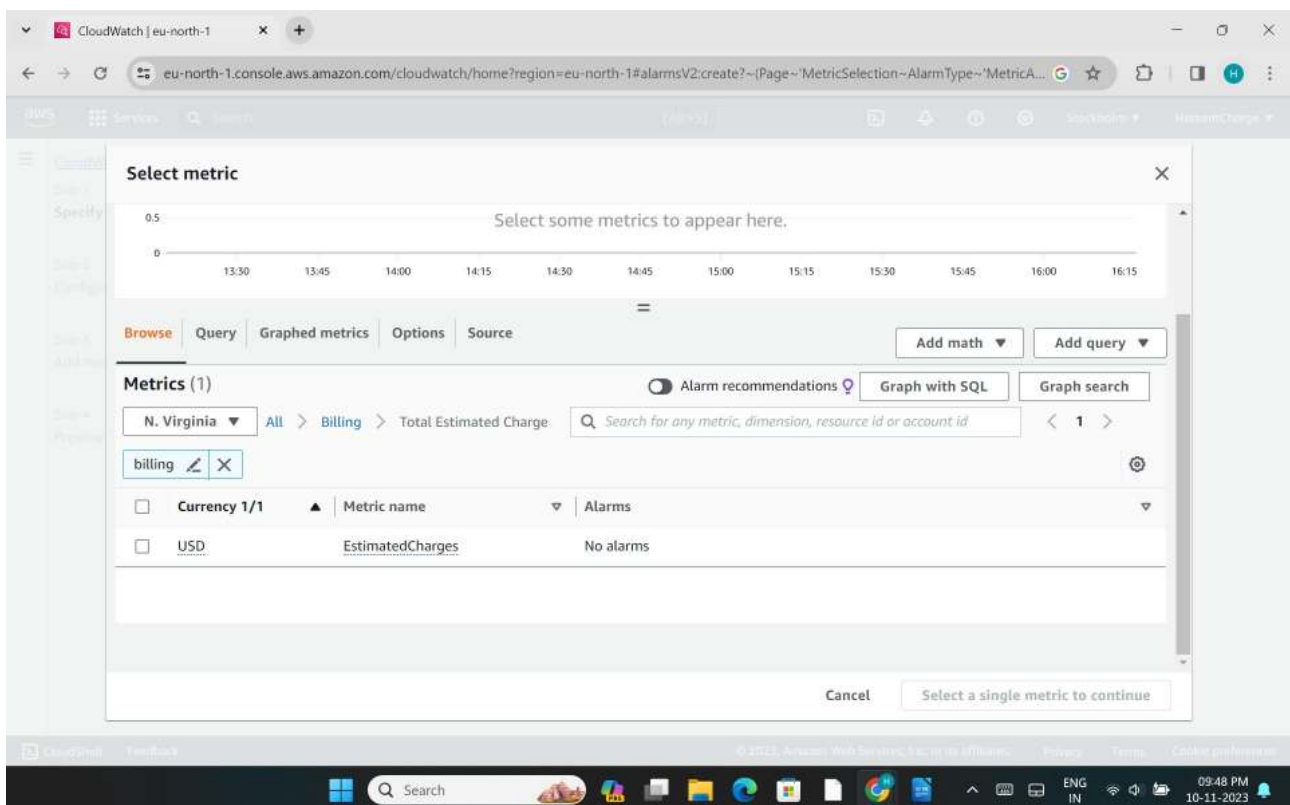
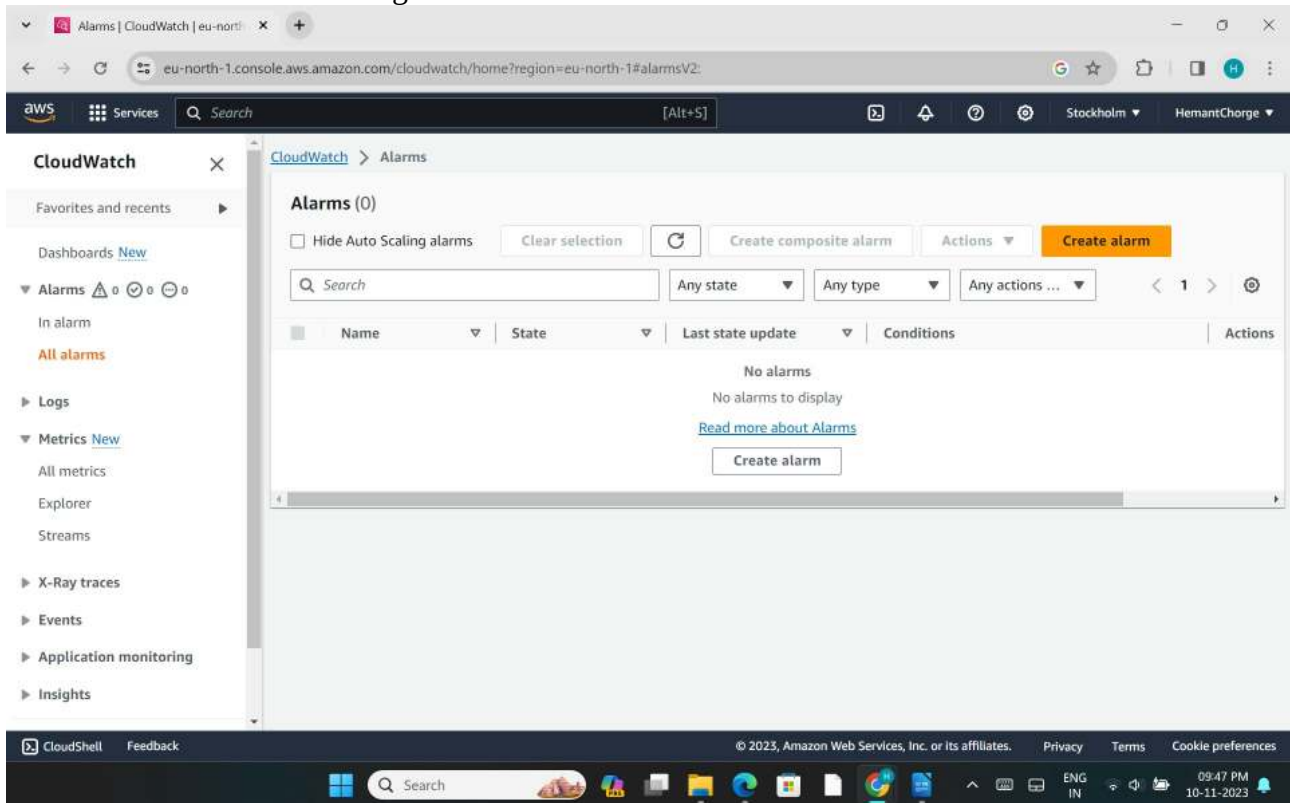
Subscription Principal
arn:aws:iam::032658000213:root

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ENG IN 09:18 PM 10-11-2023

2. Create a CloudWatch Billing Alarm



CloudWatch | eu-north-1

eu-north-1.console.aws.amazon.com/cloudwatch/home?region=eu-north-1#alarmsV2:create?~(Page~'MetricSelection~AlarmType~'MetricA...

aws Services Search [Alt+S] Stockholm HemantChorge

Conditions

Threshold type

☒ Static
Use a value as a threshold

☐ Anomaly detection
Use a band as a threshold

Whenever EstimatedCharges is...

Define the alarm condition.

☒ Greater
> threshold

☐ Greater/Equal
≥ threshold

☐ Lower/Equal
≤ threshold

☐ Lower
< threshold

than...

Define the threshold value.

500 USD

Must be a number

► Additional configuration

Cancel Next

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ENG IN 09:49 PM 10-11-2023

CloudWatch | eu-north-1

eu-north-1.console.aws.amazon.com/cloudwatch/home?region=eu-north-1#alarmsV2:create?~(Page~'Actions~AlarmType~'MetricAlarm~AL...

aws Services Search [Alt+S] Stockholm HemantChorge

CloudWatch > Alarms > Create alarm

Step 1
[Specify metric and conditions](#)

Step 2
Configure actions

Step 3
Add name and description

Step 4
Preview and create

Configure actions

Notification

Alarm state trigger
Define the alarm state that will trigger this action.

☒ In alarm
The metric or expression is outside of the defined threshold.

☐ OK
The metric or expression is within the defined threshold.

☐ Insufficient data
The alarm has just started or not enough data is available.

Remove

Send a notification to the following SNS topic
Define the SNS (Simple Notification Service) topic that will receive the notification.

☒ Select an existing SNS topic

☐ Create new topic

☐ Use topic ARN to notify other accounts

Send a notification to...

billingalert

Only email lists for this account are available.

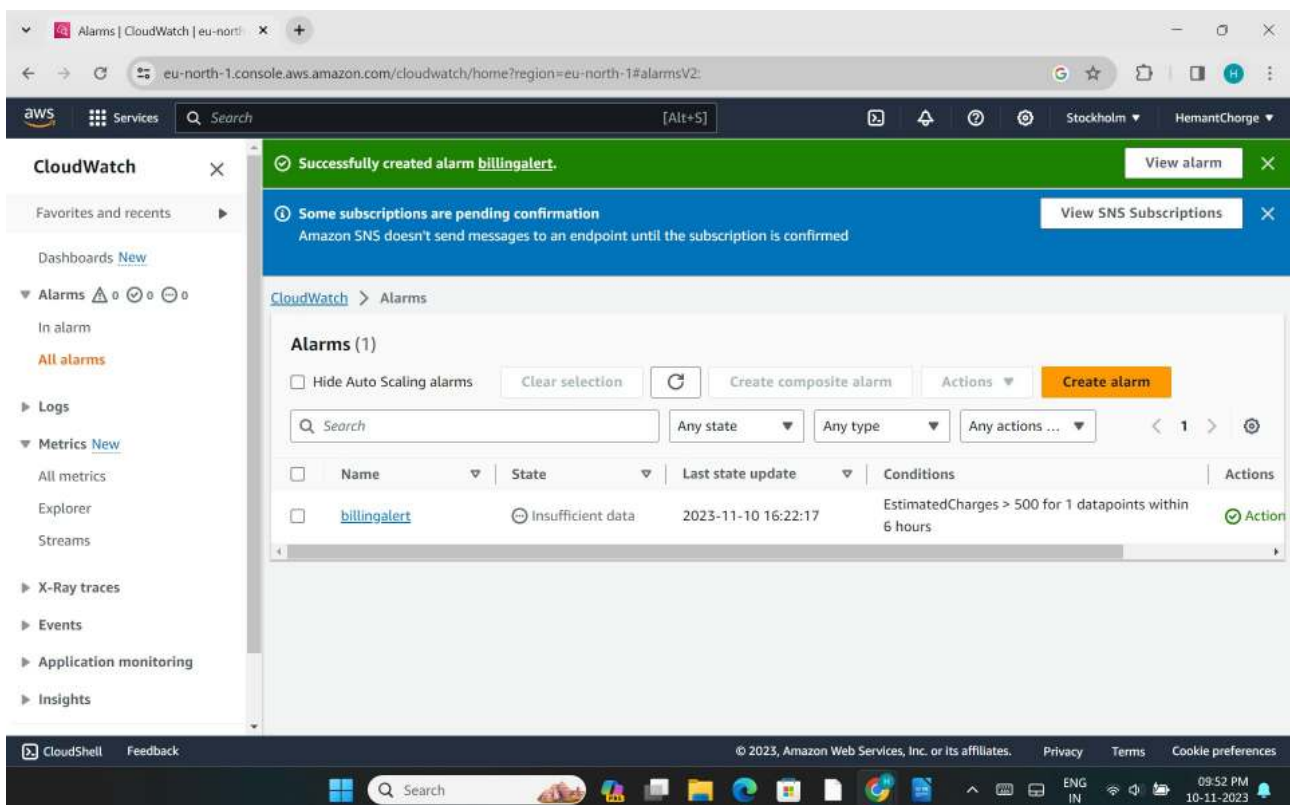
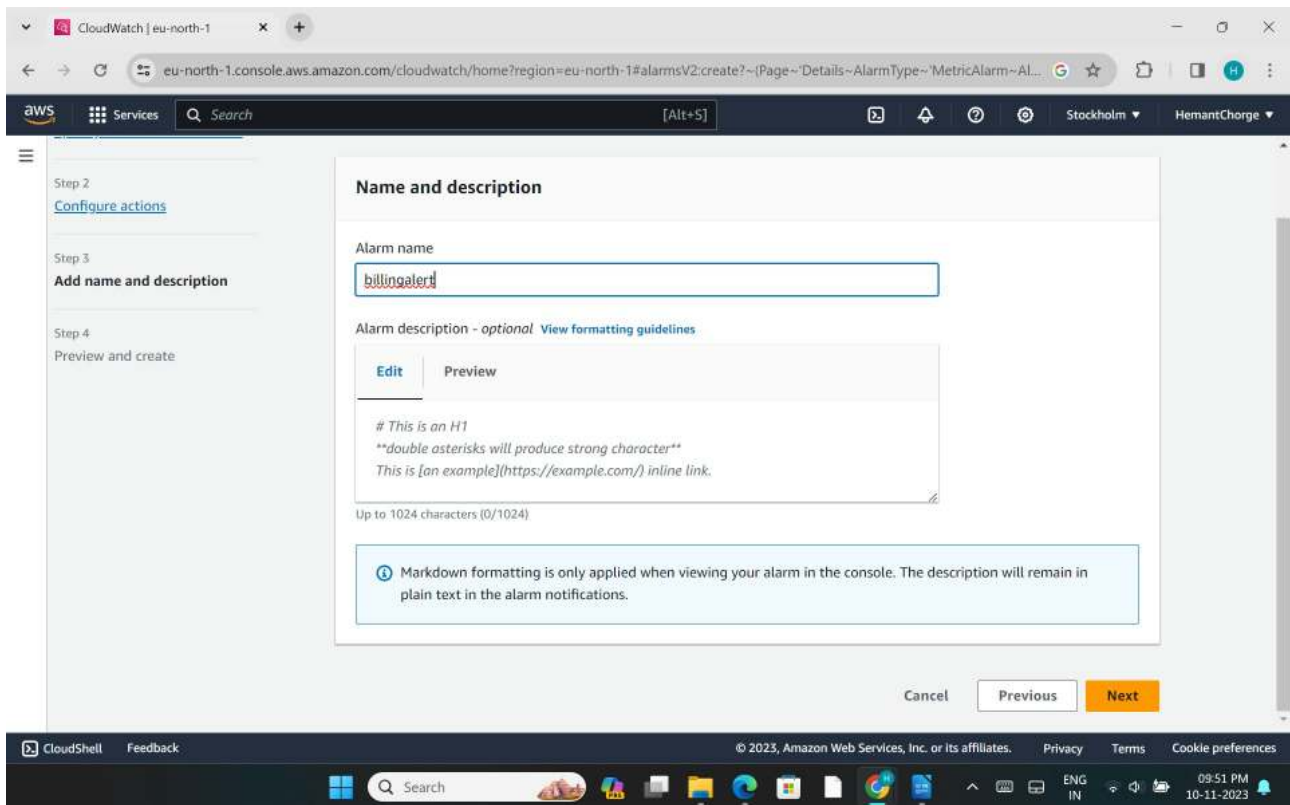
Email (endpoints)
hemantchorge45@outlook.com - View in SNS Console

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ENG IN 09:50 PM 10-11-2023



3. Create a CloudWatch Alarm for EC2 CPU Utilization

The screenshot shows the AWS CloudWatch console in the 'eu-north-1' region. The page title is 'Specify metric and conditions'. On the left, there are four steps: Step 1 (Specify metric and conditions), Step 2 (Configure actions), Step 3 (Add name and description), and Step 4 (Preview and create). The main area displays a graph for 'CPUUtilization' with a blue line and a red threshold line. The graph shows a peak in CPU utilization around 14:30. The right sidebar contains the following fields: Namespace (AWS/EC2), Metric name (CPUUtilization), InstanceId (i-097e7a86f8f1ba230), Instance name (No name specified), and Statistic (Average).

Specify metric and conditions

Graph
This alarm will trigger when the blue line goes above the red line for 1 datapoints within 5 minutes.

No unit
1
0.5
0
13:30 14:30 15:30
CPUUtilization

Namespace
AWS/EC2

Metric name
CPUUtilization

InstanceId
i-097e7a86f8f1ba230

Instance name
No name specified

Statistic
Average

The screenshot shows the AWS CloudWatch console in the 'eu-north-1' region. The page title is 'Configure actions'. On the left, there are four steps: Step 1 (Specify metric and conditions), Step 2 (Configure actions), Step 3 (Add name and description), and Step 4 (Preview and create). The main area displays a graph for 'CPUUtilization' with a blue line and a red threshold line. The graph shows a peak in CPU utilization around 14:30. The right sidebar contains the following fields: Namespace (AWS/EC2), Metric name (CPUUtilization), InstanceId (i-0c911d4520b61cd04), Instance name (demoinstance), Statistic (Maximum), and Period (5 minutes).

Configure actions

Graph
This alarm will trigger when the blue line goes above the red line for 1 datapoints within 5 minutes.

Percent
65
32.5
1e-3
13:30 14:30 15:30
CPUUtilization

Namespace
AWS/EC2

Metric name
CPUUtilization

InstanceId
i-0c911d4520b61cd04

Instance name
demoinstance

Statistic
Maximum

Period
5 minutes

CloudWatch | eu-north-1 x Instance details | EC2 | eu-north-1 x +

eu-north-1.console.aws.amazon.com/cloudwatch/home?region=eu-north-1#alarmsV2:create?~(Page~'MetricSelection~AlarmType~'MetricA...

aws Services Search [Alt+S] Stockholm HemantChorge

Conditions

Threshold type:

☒ Static
Use a value as a threshold

☐ Anomaly detection
Use a band as a threshold

Whenever CPUUtilization is...
Define the alarm condition.

☒ Greater
> threshold

☐ Greater/Equal
≥ threshold

☐ Lower/Equal
≤ threshold

☐ Lower
< threshold

than...
Define the threshold value.

65

Must be a number

► Additional configuration

Cancel Next

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09:58 PM 10-11-2023

CloudWatch | eu-north-1 x Instance details | EC2 | eu-north-1 x +

eu-north-1.console.aws.amazon.com/cloudwatch/home?region=eu-north-1#alarmsV2:create?~(Page~'Actions~AlarmType~'MetricAlarm~AL...

aws Services Search [Alt+S] Stockholm HemantChorge

Configure actions

Step 3
Add name and description

Step 4
Preview and create

Notification

Alarm state trigger
Define the alarm state that will trigger this action.

☒ In alarm
The metric or expression is outside of the defined threshold.

☐ OK
The metric or expression is within the defined threshold.

☐ Insufficient data
The alarm has just started or not enough data is available.

Remove

Send a notification to the following SNS topic
Define the SNS (Simple Notification Service) topic that will receive the notification.

☒ Select an existing SNS topic

☐ Create new topic

☐ Use topic ARN to notify other accounts

Send a notification to...

billingalert

Only email lists for this account are available.

Email (endpoints)
hemantchorge45@outlook.com - View in SNS Console

Add notification

Auto Scaling action

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09:59 PM 10-11-2023

CloudWatch | eu-north-1 x Instance details | EC2 | eu-north-1 x +

eu-north-1.console.aws.amazon.com/cloudwatch/home?region=eu-north-1#alarmsV2:create?~(Page~Details~AlarmType~MetricAlarm~Al...

aws Services Search [Alt+S] Stockholm HemantChorge

Step 2
[Configure actions](#)

Step 3
Add name and description

Step 4
Preview and create

Name and description

Alarm name
ec2cpu

Alarm description - optional [View formatting guidelines](#)

Edit Preview

This is an H1
double asterisks will produce strong character
This is [an example](https://example.com/) inline link.

Up to 1024 characters (0/1024)

Markdown formatting is only applied when viewing your alarm in the console. The description will remain in plain text in the alarm notifications.

Cancel Previous Next

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10:00 PM 10-11-2023

CloudWatch | eu-north-1 x Instance details | EC2 | eu-north-1 x +

eu-north-1.console.aws.amazon.com/cloudwatch/home?region=eu-north-1#alarmsV2:create?~(Page~Preview~AlarmType~MetricAlarm~Al...

aws Services Search [Alt+S] Stockholm HemantChorge

Step 2: Configure actions Edit

Actions

Notification
When In alarm, send a notification to "billingalert"

Step 3: Add name and description Edit

Name and description

Name
ec2cpu

Description
-

Cancel Previous Create alarm

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Search

10:00 PM 10-11-2023

4. All alarms:

The screenshot displays the AWS CloudWatch Alarms console in the eu-north-1 region. The left sidebar shows the navigation menu with options like Dashboards, Alarms, Logs, Metrics, X-Ray traces, Events, Application monitoring, and Insights. The main content area shows a notification bar at the top with a blue message about pending SNS subscriptions and a green message confirming the successful creation of the 'ec2cpu' alarm. Below this, the 'Alarms (2)' section is visible, featuring a table with two alarms:

Name	State	Last state update	Conditions	Actions
ec2cpu	Insufficient data	2023-11-10 16:30:23	CPUUtilization > 65 for 1 datapoints within 5 minutes	Action
billingalert	Insufficient data	2023-11-10 16:22:17	EstimatedCharges > 500 for 1 datapoints within 6 hours	Action

The bottom of the screenshot shows the Windows taskbar with the time 10:00 PM on 10-11-2023.